

INTERNATIONAL BACCALAUREATE  
**Mathematics: analysis and approaches**  
**MAA**

**EXERCISES [MAA 3.4]**  
**ARCS AND SECTORS**  
*Compiled by Christos Nikolaidis*

**O. Practice questions**

1. [Maximum mark: 6] **[without GDC]**

(a) Express the following angles in radians; give your answer in terms of  $\pi$ .

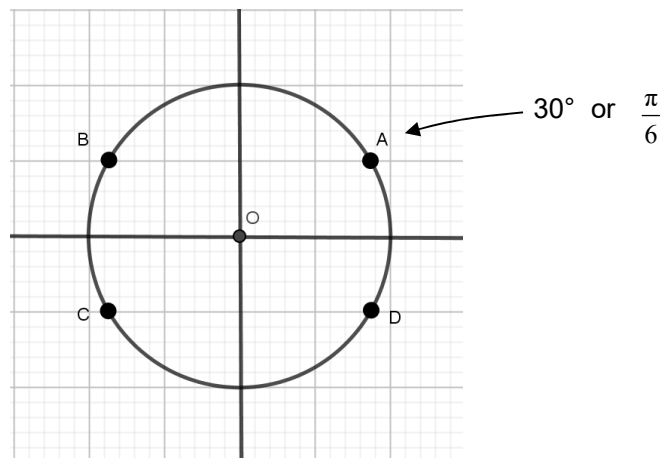
(i)  $20^\circ$       (ii)  $18^\circ$       (iii)  $540^\circ$  [3]

(b) Express the following angles in degrees

(i)  $\frac{\pi}{18}$  rad      (ii)  $\frac{\pi}{5}$  rad      (iii)  $2.5\pi$  rad. [3]

2. [Maximum mark: 12] **[without GDC]**

In the diagram below, the point A represents the angle of  $30^\circ$ , or otherwise  $\frac{\pi}{6}$  rad, on the unit circle.



(a) Complete the table below with the angle values corresponding to the symmetric points B, C and D,

(i) in degrees within the interval  $[0^\circ, 360^\circ]$

(ii) in radians within the interval  $[0, 2\pi]$  [6]

|                   | A               | B | C | D |
|-------------------|-----------------|---|---|---|
| <b>in degrees</b> | $30^\circ$      |   |   |   |
| <b>in radians</b> | $\frac{\pi}{6}$ |   |   |   |

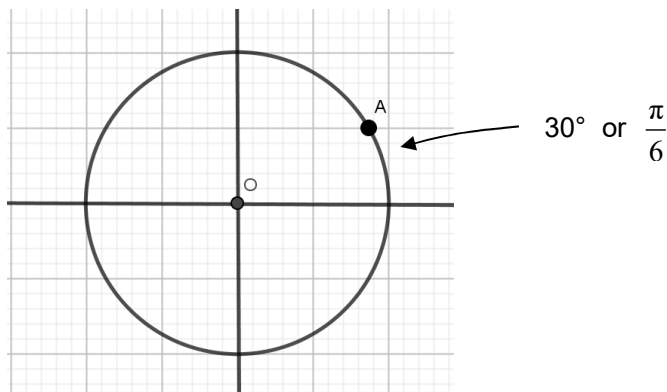
Suppose now that C represents the angle of  $220^\circ$ , or otherwise  $\frac{11\pi}{9}$  rad, in the third quadrant. Let A, B and D be the symmetric points of C on the unit circle, as shown in the diagram above.

(b) Complete the table below in a similar way as in (a). [6]

|            | A | B | C                 | D |
|------------|---|---|-------------------|---|
| in degrees |   |   | $220^\circ$       |   |
| in radians |   |   | $\frac{11\pi}{9}$ |   |

3. [Maximum mark: 8] [without GDC]

In the diagram below, the point A represents the angle of  $30^\circ$ , or otherwise  $\frac{\pi}{6}$  rad, on the unit circle.



The general formula for the angles corresponding to point A is

$$\begin{array}{ll} \text{in degrees:} & 30^\circ + 360^\circ k \quad k \in \mathbb{Z} \\ \text{in radians:} & \frac{\pi}{6} + 2k\pi \quad k \in \mathbb{Z} \end{array}$$

Determine the values of the angle at point A within the following intervals:

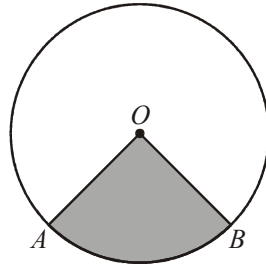
|                                  | In degrees                            |            |
|----------------------------------|---------------------------------------|------------|
| 2 <sup>nd</sup> period backwards | $-720^\circ \leq \theta < -360^\circ$ |            |
| 1 <sup>st</sup> period backwards | $-360^\circ \leq \theta < 0^\circ$    |            |
| 1 <sup>st</sup> period           | $0^\circ \leq \theta < 360^\circ$     | $30^\circ$ |
| 2 <sup>nd</sup> period           | $360^\circ \leq \theta < 720^\circ$   |            |
| 3 <sup>rd</sup> period           | $720^\circ \leq \theta < 1080^\circ$  |            |

| in radians                  |                 |
|-----------------------------|-----------------|
| $-4\pi \leq \theta < -2\pi$ |                 |
| $-2\pi \leq \theta < 0$     |                 |
| $0 \leq \theta < 2\pi$      | $\frac{\pi}{6}$ |
| $2\pi \leq \theta < 4\pi$   |                 |
| $4\pi \leq \theta < 6\pi$   |                 |

[8]

4. [Maximum mark: 20] **[without GDC]**

The diagram shows a circle center  $O$  radius of 10 cm.



The minor sector is shaded while the major sector is unshaded.

(a) Given that the size of  $\angle AOB$  is 1.5 rad, complete the table.

[10]

|                        |                          |  |
|------------------------|--------------------------|--|
| <b>Length</b><br>of    | the <b>minor</b> arc AB  |  |
|                        | the <b>major</b> arc AB  |  |
| <b>Area</b><br>of      | the <b>minor</b> sector  |  |
|                        | the <b>major</b> sector. |  |
| <b>Perimeter</b><br>of | the <b>minor</b> sector  |  |
|                        | the <b>major</b> sector  |  |

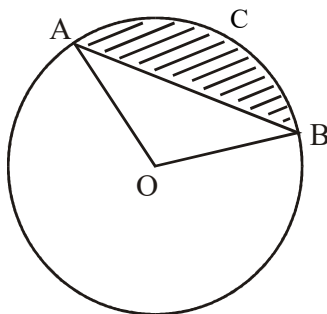
(b) Given that the size of  $\angle AOB$  is  $\frac{\pi}{2}$  rad, complete the table.

[10]

|                        |                          |  |
|------------------------|--------------------------|--|
| <b>Length</b><br>of    | the <b>minor</b> arc AB  |  |
|                        | the <b>major</b> arc AB  |  |
| <b>Area</b><br>of      | the <b>minor</b> sector  |  |
|                        | the <b>major</b> sector. |  |
| <b>Perimeter</b><br>of | the <b>minor</b> sector  |  |
|                        | the <b>major</b> sector  |  |

5. [Maximum mark: 11] **[with GDC]**

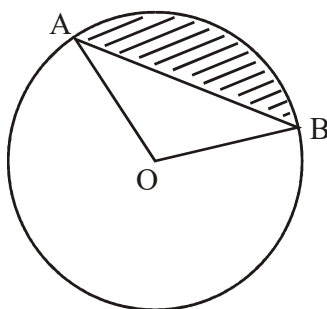
The diagram shows a circle centre  $O$  and radius  $r = 8$ . The angle  $\hat{AOB}$  is  $\theta = 1.3$  radians.



- (a) Find the length of the arc  $ACB$ . [2]
- (b) Find the area of the sector  $OACB$ . [2]
- (c) Find the length of the chord  $AB$ . [3]
- (d) Find the area of the triangle  $OAB$ . [2]
- (e) Find the area of the shaded segment. [2]

6\*. [Maximum mark: 14] **[with GDC]**

The diagram shows a circle centre  $O$  and radius  $r$ . The angle  $\hat{AOB}$  is  $\theta$  radians.



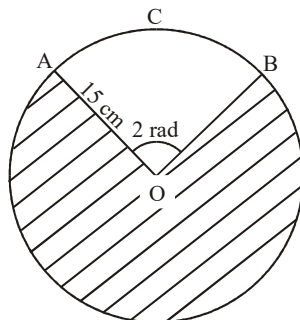
Find the value of  $\theta$ , in each of the following three cases

- (a) Given that the area of the triangle  $OAB$  is equal to the shaded area. [4]
- (b) Given that ratio of the lengths of the two arcs (minor  $AB$  to major  $AB$ ) is 1:5. [4]
- (c) Given that the ratio of the shaded area to the non-shaded area is 1:5. [5]

**A. Exam style questions (SHORT)**

**7. [Maximum mark: 6] [with GDC]**

The following diagram shows a circle of centre  $O$ , and radius  $15\text{ cm}$ . The arc  $ACB$  subtends an angle of  $2$  radians at the centre  $O$ .



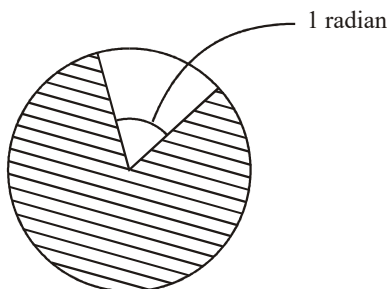
**Diagram not to scale**

$$\begin{aligned} \angle AOB &= 2 \text{ radians} \\ OA &= 15 \text{ cm} \end{aligned}$$

- (a) Find the length of the arc  $ACB$ ; [2]  
 (b) Find the area of the shaded region. [4]

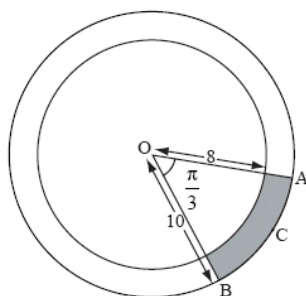
**8. [Maximum mark: 4] [with / without GDC]**

The diagram shows a circle of radius  $5\text{ cm}$ . Find the perimeter of the shaded region.



**9. [Maximum mark: 6] [without GDC]**

The diagram shows two concentric circles with centre  $O$ .

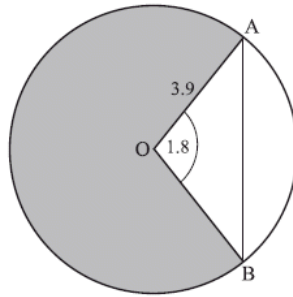


The radius of the smaller circle is  $8\text{ cm}$ , the radius of the larger circle is  $10\text{ cm}$ . Points  $A$ ,  $B$  and  $C$  are on the circumference of the larger circle such that  $\angle AOB$  is  $\frac{\pi}{3}$  radians

- (a) Find the length of the arc  $ACB$ . [2]  
 (b) Find the area of the shaded region. [4]

10. [Maximum mark: 7] **[with GDC]**

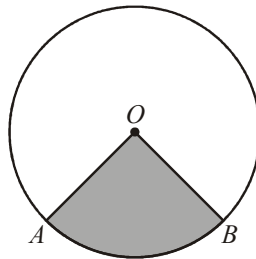
The circle shown has centre  $O$  and radius 3.9 cm. Points  $A$  and  $B$  lie on the circle and angle  $AOB$  is 1.8 radians.



- (a) Find  $AB$ . [3]  
 (b) Find the area of the shaded region. [4]

11. [Maximum mark: 4] **[with GDC]**

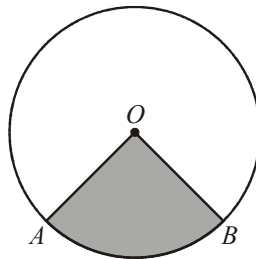
$O$  is the centre of the circle which has a radius of 5.4 cm.



The area of the shaded sector  $OAB$  is  $21.6 \text{ cm}^2$ . Find the length of the minor arc  $AB$ .

12. [Maximum mark: 6] **[with GDC]** [diagram as above]

The diagram below shows a circle of centre  $O$ , and radius  $r$ .

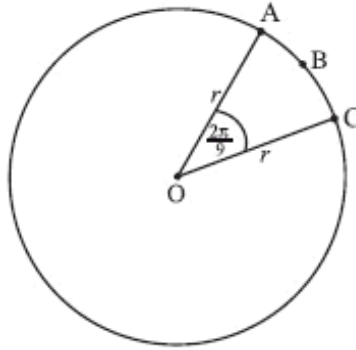


The shaded sector  $OAB$  has an area of  $27 \text{ cm}^2$ . Angle  $\hat{AOB} = \theta = 1.5$  radians.

- (a) Find the radius  $r$ . [4]  
 (b) Calculate the length of the minor arc  $AB$ . [2]

13. [Maximum mark: 6] **[with GDC]**

The diagram below shows a circle centre O, with radius  $r$ .

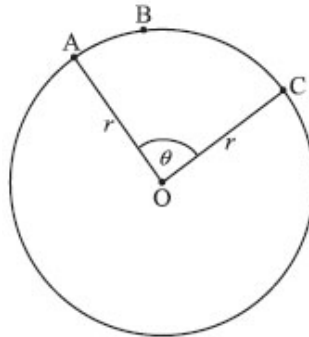


The length of arc ABC is  $3\pi$  cm and  $\angle AOC = \frac{2\pi}{9}$ .

- (a) Find the value of  $r$ . [2]
- (b) Find the perimeter of sector OABC. [2]
- (c) Find the area of sector OABC. [2]

14. [Maximum mark: 6] **[without GDC]**

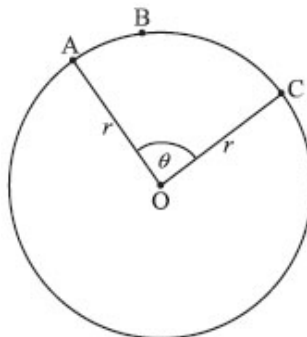
The following diagram shows a circle with radius  $r$  and centre O. The points A, B and C are on the circle and  $\angle AOC = \theta$ .



The area of sector OABC is  $\frac{4}{3}\pi$ , the length of arc ABC is  $\frac{2}{3}\pi$ .

15. [Maximum mark: 6] **[with GDC]**

The area of the sector OAC is  $180 \text{ cm}^2$ , the length of the arc AC is 24 cm.



Find the value of  $r$  and of  $\theta$ .

16. [Maximum mark: 5] **[with GDC]**

The diagram below shows a sector AOB of a circle of radius 15 cm and centre O. The angle  $\theta$  at the centre of the circle is 2 radians.

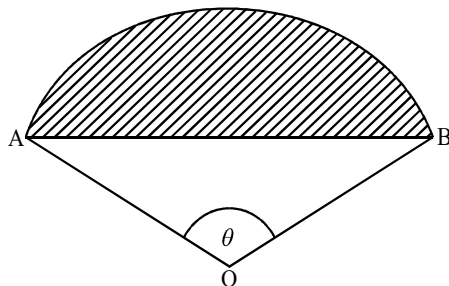
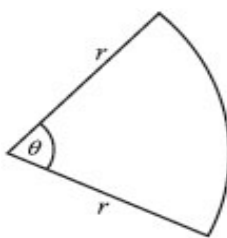


Diagram not to scale

- (a) Calculate the area of the sector AOB. [2]  
 (b) Calculate the area of the shaded region. [3]

17. [Maximum mark: 6] **[with GDC]**

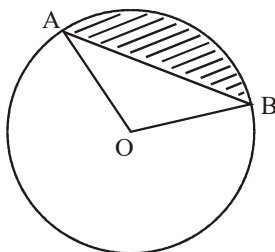
The following diagram shows a sector of a circle of radius  $r$  cm, and angle  $\theta$  at the centre. The perimeter of the sector is 20 cm.



- (a) Show that  $\theta = \frac{20-2r}{r}$ . [2]  
 (b) The area of the sector is  $25 \text{ cm}^2$ . Find the value of  $r$ . [4]

18. [Maximum mark: 6] **[with GDC]**

The diagram shows a circle centre O and radius  $OA=5 \text{ cm}$ . The angle  $AOB = 135^\circ$ .

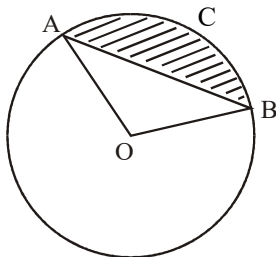


Find the area of the shaded region.



19. [Maximum mark: 6] **[with GDC]**

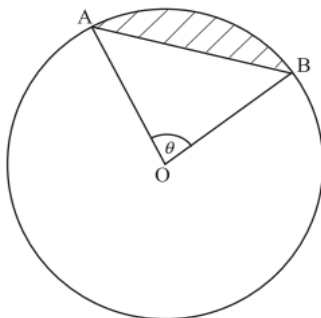
The diagram shows a circle centre  $O$  and radius  $r$ . The length of the arc  $ACB$  is  $2r$ .



The area of the shaded segment may be expressed as  $kr^2$ . Find the value of  $k$ .

20\*. [Maximum mark: 6] **[with GDC]**

The diagram shows a circle centre  $O$  and radius 1, with  $\angle AOB = \theta$ ,  $\theta \neq 0$ . The area of  $\triangle AOB$  is three times the shaded area.



- (a) Show that  $3\theta = 4 \sin \theta$ . [4]  
 (b) Find the value of  $\theta$ . [2]

21\*. [Maximum mark: 6] **[without GDC]**

The diagram above [Ex. 20] shows a circle centre  $O$  and radius  $r$ , with  $\angle AOB = \theta$ .

The ratio of the shaded area to the area of  $\triangle AOB$  is 2:5. Show that  $5\theta = 7 \sin \theta$ .

22\*. [Maximum mark: 6] **[with / without GDC]**

In the following diagram,  $O$  is the centre of the circle and  $(AT)$  is the tangent to the circle at  $T$ .

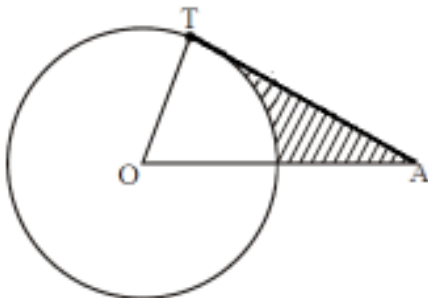
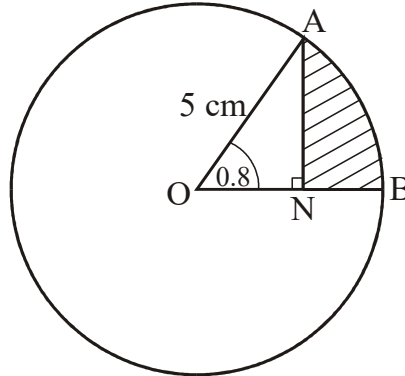


Diagram not to scale

If  $OA = 12$  cm, and the circle has a radius of 6 cm, find the area of the shaded region.

**23\*.** [Maximum mark: 6] **[with GDC]**

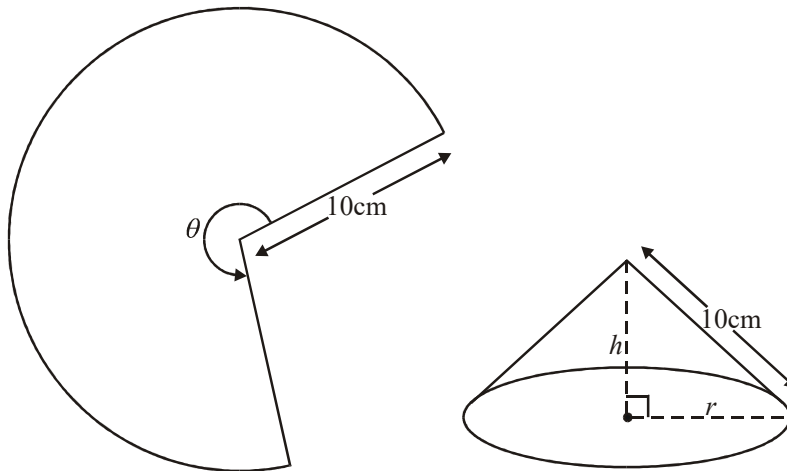
The diagram below shows a circle of radius 5 cm with centre O. Points A and B are on the circle, and  $\angle AOB$  is 0.8 radians. The point N is on [OB] such that [AN] is perpendicular to [OB]..



Find the area of the shaded region.

**24\*.** [Maximum mark: 6] **[without GDC]**

The diagrams show a circular sector of radius 10 cm and angle  $\theta$  radians which is formed into a cone of slant height 10 cm. The vertical height  $h$  of the cone is equal to the radius  $r$  of its base.



- (a) Find the value of  $r$ . [2]
- (b) Find the angle  $\theta$  radians. [4]

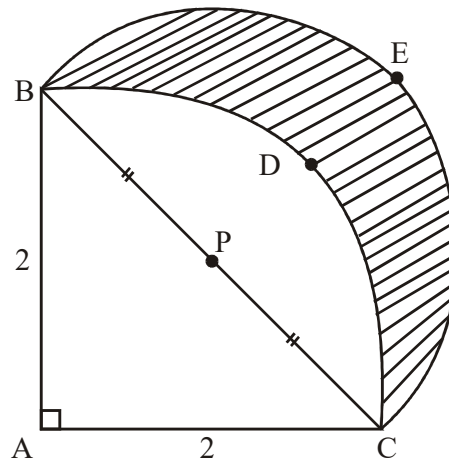
25. [Maximum mark: 6] **[without GDC]**

The diagram below shows a triangle and two arcs of circles.

The triangle ABC is a right-angled isosceles triangle, with  $AB = AC = 2$ . The point P is the midpoint of [BC].

The arc BDC is part of a circle with centre A.

The arc BEC is part of a circle with centre P.

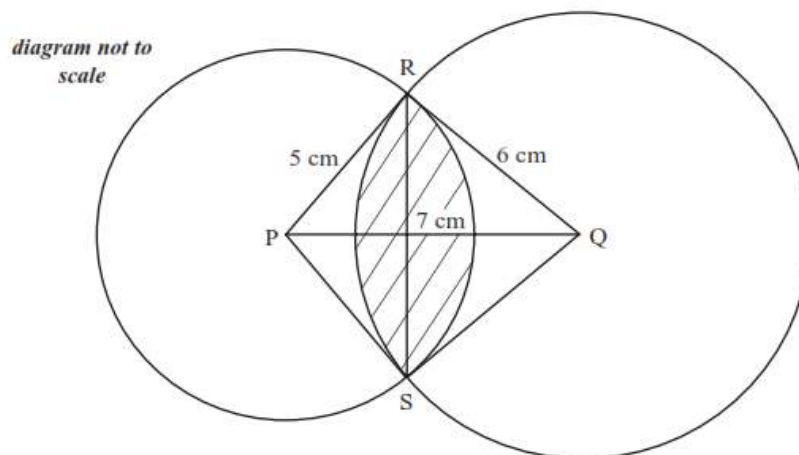


(a) Calculate the area of the segment BDCP. [3]

(b) Calculate the area of the shaded region BECD. [3]

26\*. [Maximum mark: 7] **[with GDC]**

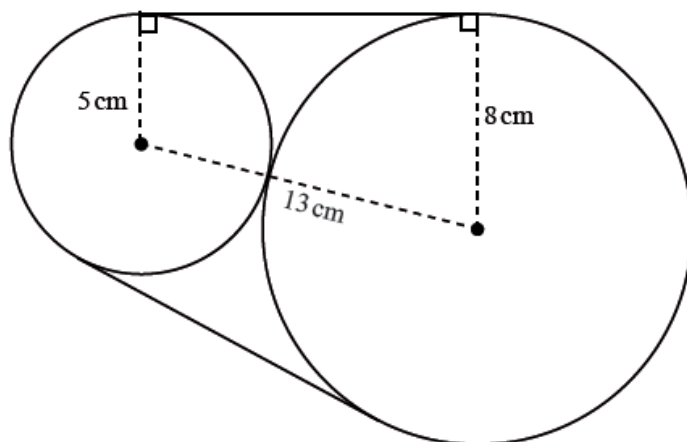
The diagram below shows a pair of intersecting circles with centres at P and Q with radii of 5 cm and 6 cm respectively. RS is the common chord of both circles and PQ is 7 cm.



Find the area of the shaded region.

**27\*.** [Maximum mark: 8] **[with GDC]**

Two discs, one of radius 8 cm and one of radius 5 cm, are placed such that they touch each other. A piece of string is wrapped around the discs. This is shown in the diagram below.



Calculate the length of the string needed to go around the discs.

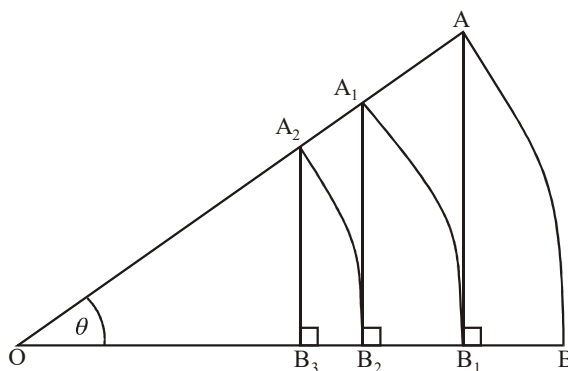
**28\*.** [Maximum mark: 5] **[without GDC]**

A circular disc is cut into twelve sectors whose areas are in an arithmetic sequence.

- Explain why the angles of the sectors are also in an arithmetic sequence. [2]
- The angle of the largest sector is twice the angle of the smallest sector. Find the size of the angle of the smallest sector. [5]

**29\*.** [Maximum mark: 7] **[with GDC]**

The diagram shows a sector AOB of a circle of radius 1 and centre O, where  $\angle AOB = \theta$ . The lines  $(AB_1)$ ,  $(A_1B_2)$ ,  $(A_2B_3)$  are perpendicular to OB.  $A_1B_1$ ,  $A_2B_2$  are all arcs of circles with centre O.

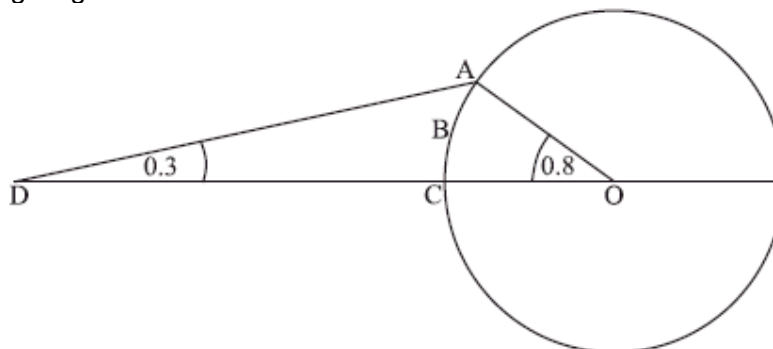


- Calculate, in terms of  $\theta$ , the sum to infinity of the arc lengths  $AB + A_1B_1 + A_2B_2 + A_3B_3 + \dots$  [5]
- Given that the sum to infinity of the arc lengths is  $\frac{2\pi}{3}$ , find the value of  $\theta$ . [2]

**B. Exam style questions (LONG)**

**30.** [Maximum mark: 13] **[with GDC]**

The following diagram shows a circle with centre O and radius 4 cm.



*diagram not to scale*

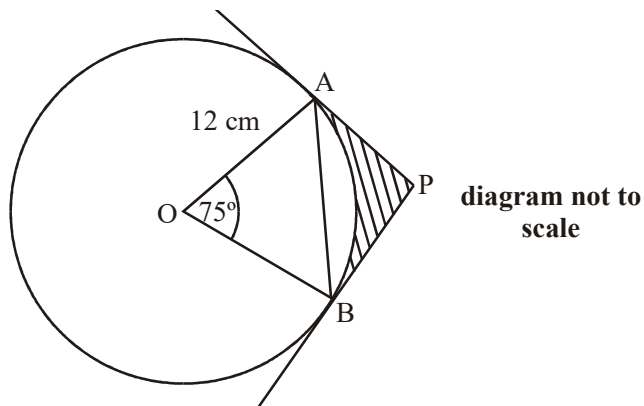
The points A, B and C lie on the circle. The point D is outside the circle, on (OC).

Angle ADC = 0.3 radians and angle AOC = 0.8 radians.

- |                                   |     |
|-----------------------------------|-----|
| (a) Find AD.                      | [3] |
| (b) Find OD.                      | [4] |
| (c) Find the area of sector OABC. | [2] |
| (d) Find the area of region ABCD. | [4] |

**31.** [Maximum mark: 13] **[with GDC]**

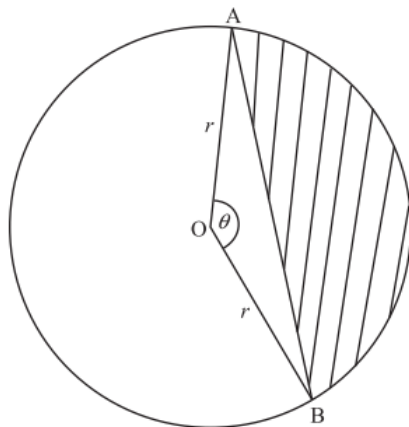
The diagram shows a circle, centre O, with a radius 12cm. The chord AB subtends at an angle of  $75^\circ$  at the centre. The tangents to the circle at A and at B meet at P.



- |                                                                                            |     |
|--------------------------------------------------------------------------------------------|-----|
| (a) Using the cosine rule, show that the length of AB is $12\sqrt{2(1 - \cos 75^\circ)}$ . | [2] |
| (b) Find the length of BP.                                                                 | [3] |
| (c) Hence find (i) the area of triangle OBP; (ii) the area of triangle ABP.                | [4] |
| (d) Find the area of <b>sector</b> OAB.                                                    | [2] |
| (e) Find the area of the shaded region.                                                    | [2] |

**32.** [Maximum mark: 10] **[with GDC]**

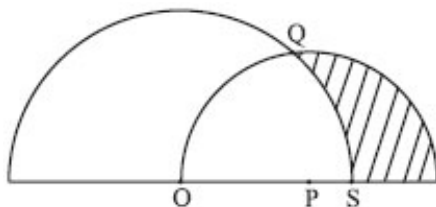
The following diagram shows a circle centre  $O$ , radius  $r$ . The angle  $\hat{AOB}$  at the centre of the circle is  $\theta$  radians. The chord  $AB$  divides the circle into a minor segment (the shaded region) and a major segment.



- (a) Show that the area of the minor segment is  $\frac{1}{2}r^2(\theta - \sin \theta)$ . [3]
- (b) Given that the ratio of the areas of the two segments is 2:3, show that  $\sin \theta = \theta - \frac{4\pi}{5}$ . [5]
- (c) Hence find the value of  $\theta$ . [2]

**33\*.** [Maximum mark: 17] **[with GDC]**

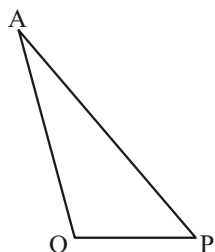
The following diagram shows two semi-circles. The larger one has centre  $O$  and radius 4 cm. The smaller one has centre  $P$ , radius 3 cm, and passes through  $O$ . The line  $(OP)$  meets the larger semi-circle at  $S$ . The semi-circles intersect at  $Q$ .



- (a)
  - (i) Explain why  $OPQ$  is an isosceles triangle.
  - (ii) Use the cosine rule to show that  $\cos \hat{OPQ} = \frac{1}{9}$ .
  - (iii) Hence show that  $\sin \hat{OPQ} = \frac{\sqrt{80}}{9}$ .
  - (iv) Find the area of the triangle  $OPQ$ . [7]
- (b) Consider the smaller semi-circle, with centre  $P$ .
  - (i) Write down the size of  $\hat{OPQ}$ .
  - (ii) Calculate the area of the sector  $OPQ$ . [3]
- (c) Consider the semi-circle, with centre  $O$ . Calculate the area of the sector  $QOS$ .
- (d) Hence calculate the area of the shaded region. [4]

**34\*.** [Maximum mark: 20] **[with GDC]**

The following diagram shows the triangle AOP, where  $OP = 2\text{cm}$ ,  $AP = 4\text{cm}$  and  $AO = 3\text{cm}$

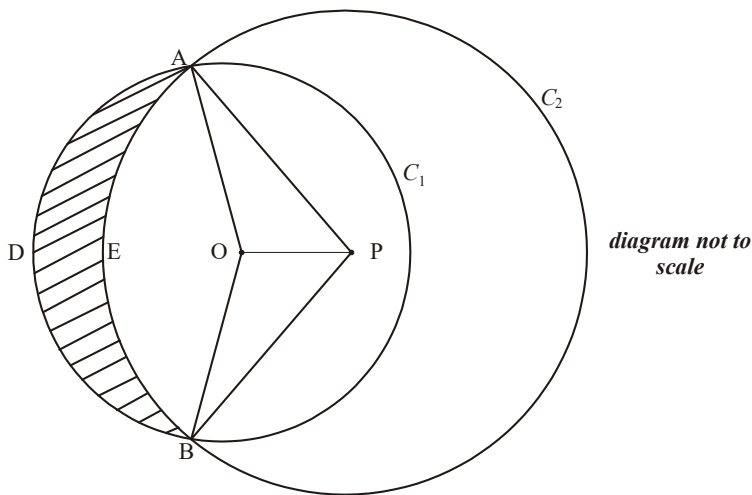


*diagram not to scale*

(a) Calculate  $\hat{AOP}$ , giving your answer in radians.

[3]

The following diagram shows two circles which intersect at the points A and B. The smaller circle  $C_1$  has centre O and radius 3 cm, the larger circle  $C_2$  has centre P and radius 4 cm, and  $OP = 2\text{ cm}$ . The point D lies on the circumference of  $C_1$  and E on the circumference of  $C_2$ . Triangle AOP is the same as triangle AOP in the diagram above.



*diagram not to scale*

(b) Find  $\hat{AOB}$ , giving your answer in radians.

[2]

(c) Show that  $\hat{APB} \cong 1.63$  radians.

[3]

(d) Calculate the area of

- (i) sector PAEB;                      (ii) sector OADB.

[5]

(e) Show that the area of the quadrilateral AOBP  $\cong 5.81\text{ cm}^2$ .

[3]

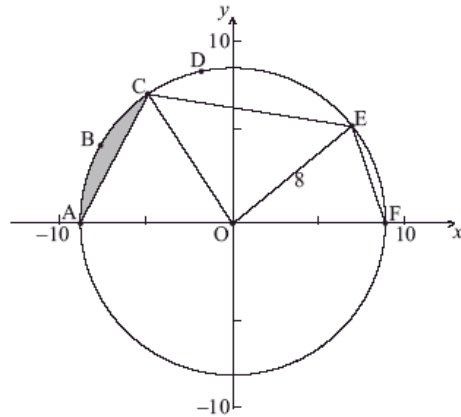
(f) Find the areas of

- (i) AOB.                                      (ii) the shaded region AEBD.

[4]

35. [Maximum mark: 15] **[with GDC]**

The diagram below shows a circle with centre O and radius 8 cm.



**diagram not to scale**

The points A, B, C, D, E and F are on the circle, and [AF] is a diameter. The length of arc ABC is 6 cm.

- (a) Find the size of angle AOC. [2]
- (b) Hence find the area of the shaded region. [6]

The area of sector OCDE is  $45 \text{ cm}^2$ .

- (c) Find the size of angle COE. [2]
- (d) Find EF. [5]